



IS, STRATEGIES & MANAGEMENT ACQUISITION

Md. Iftekhar Hossain
Assistant Professor
CSE, NBIU.

LESSON 1.1-INFORMATION SYSTEM - CONTEXTUAL

DATA:

Plural form of datum

- Raw facts and figures that on their own have no meaning
- Unorganized piece of information

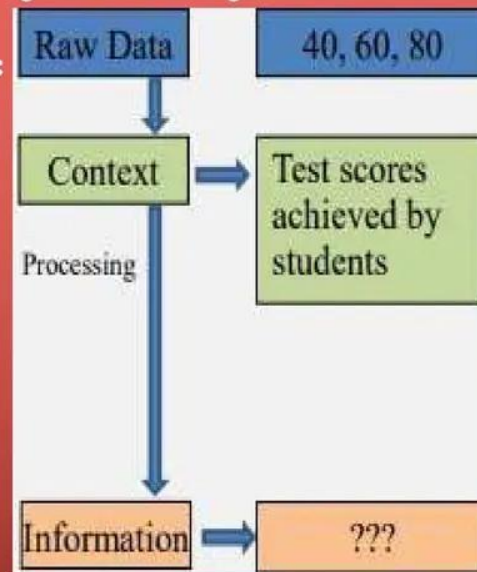
Example: 40, 60, 80,

Yes, No

INFORMATION:

Data that has been processed within a context to give it meaning.

Example:

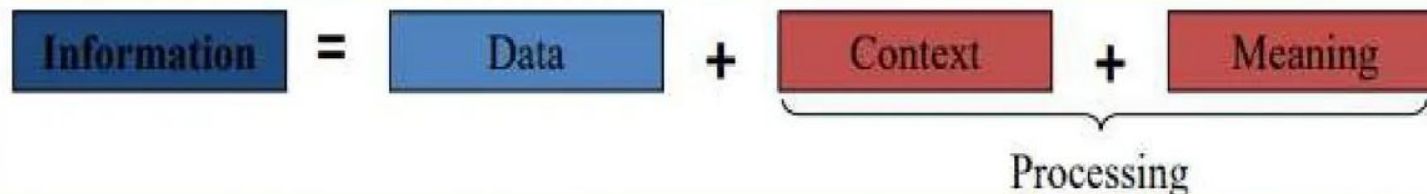


KNOWLEDGE:

- Organized body of information or capability of understanding the relationship between pieces of information and what to actually do with the information.

Example:

Average test score is 60%. This shows that the first student's marks is below average.



INFORMATION SYSTEMS (IS)

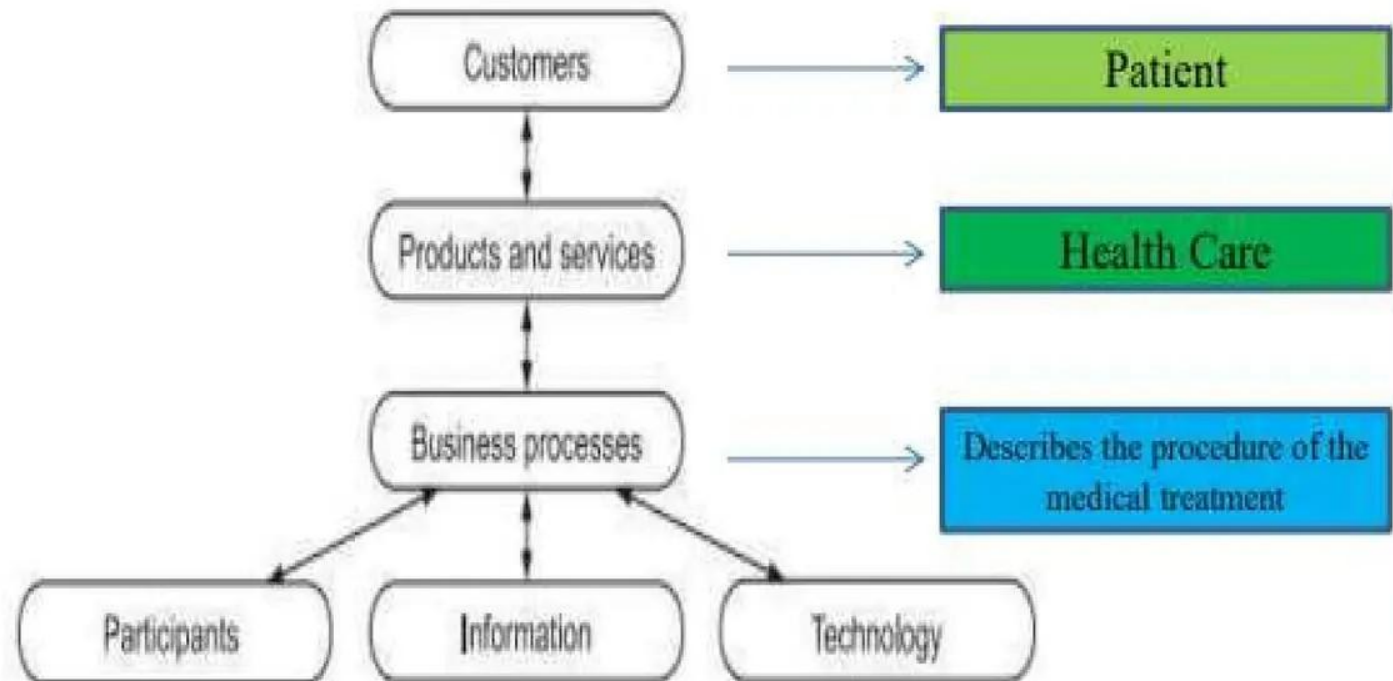
- Information = Latin words (Formation and Forma)
- An Information System (IS) is a group of components that interact to produce information.
- An Information System is an integrated and cooperating set of software directed information technologies supporting individual, group, organizational, or societal goals.
- It is the study of complementary networks that people and organizations use to collect, filter, process create and distribute data.

AN INTEGRATED VIEW OF AN INFORMATION SYSTEM

An information system is a software system to capture, transmit, store, retrieve, manipulate, or display information, thereby supporting people, organizations, or other software systems.

Example: In the example of the family doctor, the Information System is the software system that stores the data

Information System Example



TWO PERSPECTIVES OF AN INFORMATION SYSTEMS

1. Function Perspective

Information system is a technologically implemented medium for the *purpose of recording, storing, and disseminating linguistic expressions as well as for the supporting of inference making*".

Example: Communicate with experts to solve a particular problem

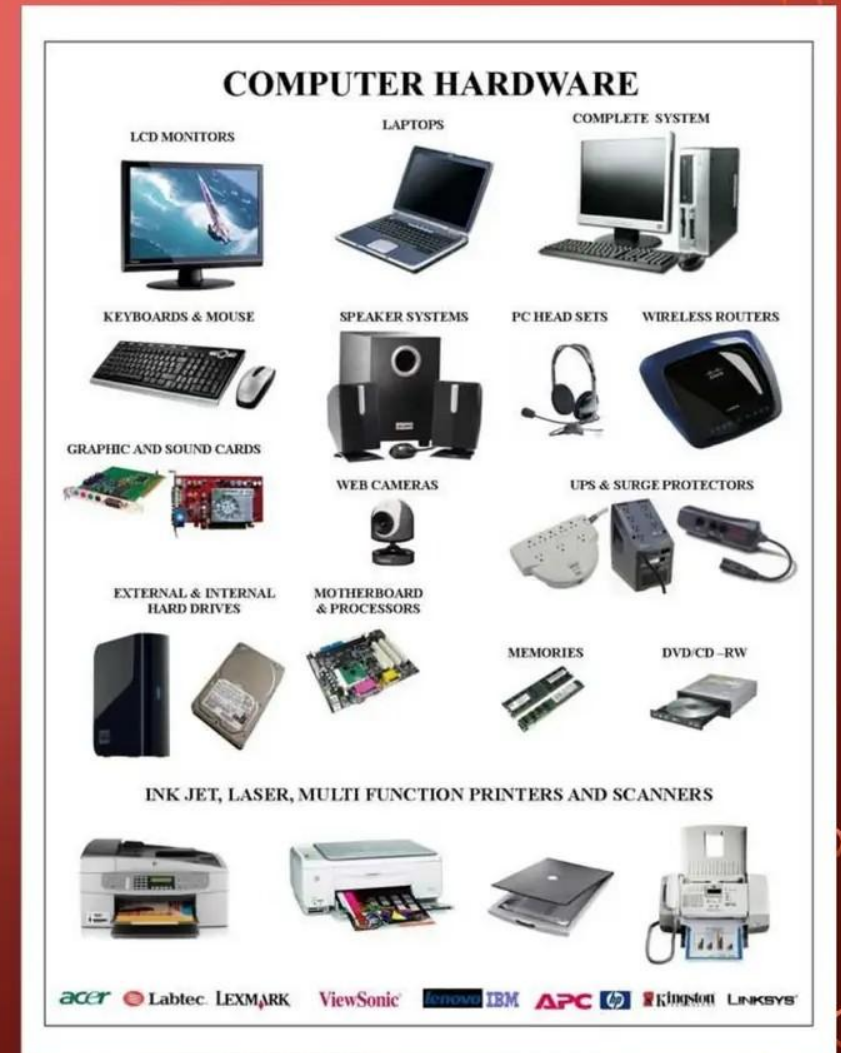
2. Structure Perspective

An information system consists of a *collection of people, processes, data, models, technology and partly formalized language, forming a cohesive structure which serves some organizational purpose or function.*

Example: Socio-technical systems, i.e., systems consisting of humans, behaviour rules, and conceptual and technical artifacts.

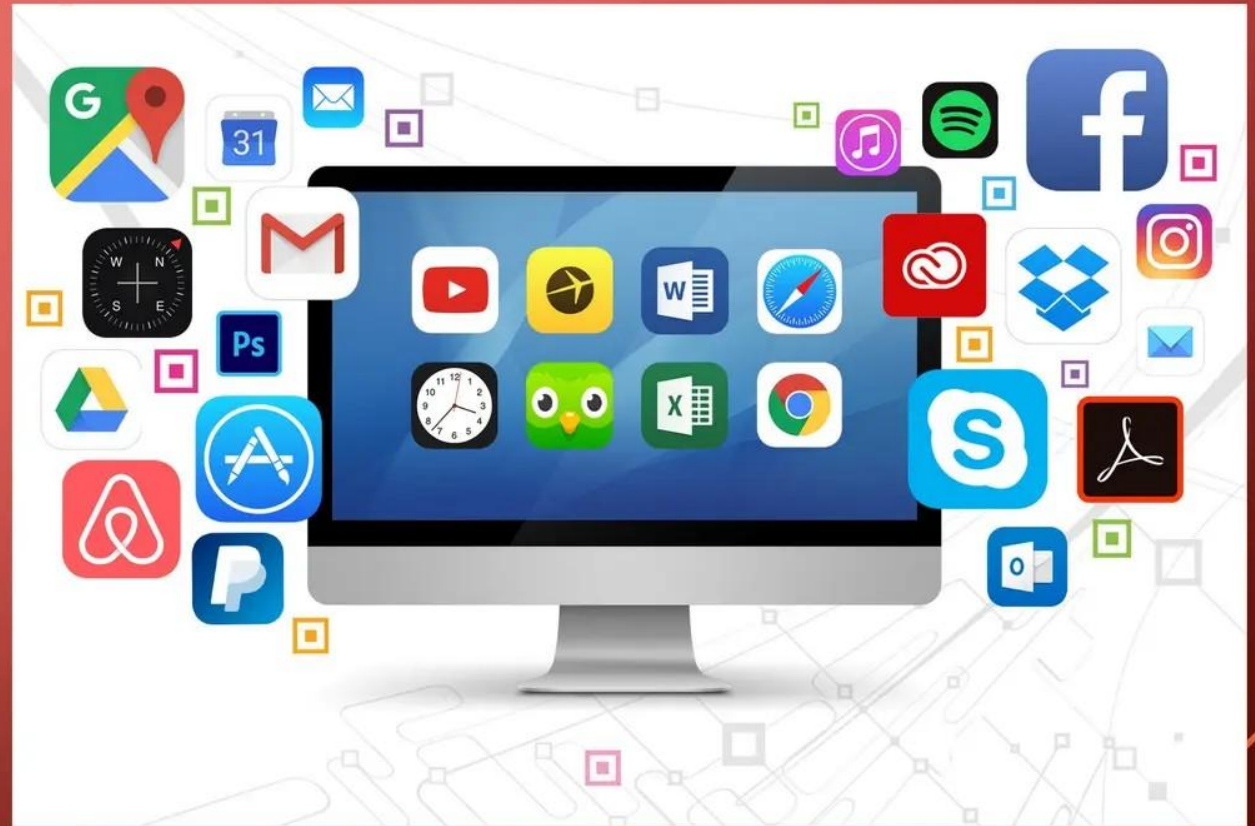
COMPONENTS OF AN INFORMATION SYSTEMS

1. *Hardware*: Physical computer equipment's and associate device, input and output devices, storage devices and communications devices.



COMPONENTS OF AN INFORMATION SYSTEMS

2. **Software:** The term software refers to computer programs and the manuals that support hardware (programs and procedures).



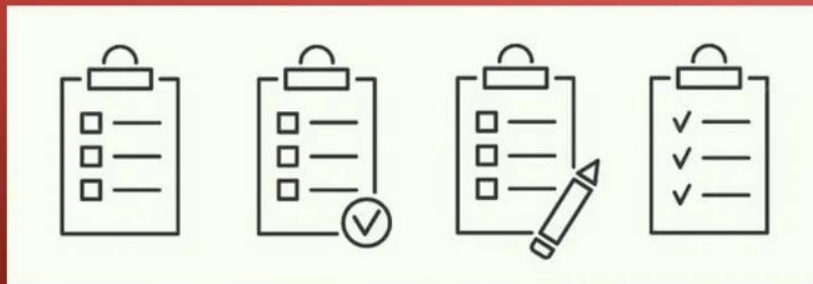
COMPONENTS OF AN INFORMATION SYSTEMS

3. *Data*: Data are facts that are used by programs to produce useful information. (Data and knowledge bases).



COMPONENTS OF AN INFORMATION SYSTEMS

4. *Procedures/process*: Procedures are the policies that govern the operation of a computer system.



COMPONENTS OF AN INFORMATION SYSTEMS

5. *People*: Every system needs people if it is to be useful (end users and IS specialists, system analyst, programmers, data administrators etc.)



COMPONENTS OF AN INFORMATION SYSTEMS

6. *Networks*: Communications media and network support (Internet, intranets, extranets, computers, communications processors, and other devices).



A decorative graphic on the left side of the slide, consisting of a network of thin, light-orange lines that resemble a circuit board or a neural network. These lines are interconnected with small, hollow orange circles of varying sizes, creating a complex, branching pattern that extends from the top to the bottom of the frame.

SYSTEMS FEATURES AND CHARACTERISTICS

SYSTEMS ALSO EXHIBIT CERTAIN FEATURES AND CHARACTERISTICS:

1. *Objective*

- Every system has a predefined goal or objective towards which it works.
- An organization would have an objective of earning maximum possible revenues.

SYSTEMS ALSO EXHIBIT CERTAIN FEATURES AND CHARACTERISTICS:

2. Standards

- Systems should be designed to meet standards.
- Standards can be business specific or organization specific.
- There should be standard rules to use a particular system.

SYSTEMS ALSO EXHIBIT CERTAIN FEATURES AND CHARACTERISTICS:

3. *Environment*

- Every system whether it is natural or man-made co-exists with an environment.

Example: Y2K problem for computer systems.

- Systems that are not Y2K compliant will not work properly after year 2000.
- It is important to make the systems Y2K compliant.

SYSTEMS ALSO EXHIBIT CERTAIN FEATURES AND CHARACTERISTICS:

4. *Feedback*

- The output of a system needs to be scrutinised by taking feedbacks for improving the system.
- Also some feedback can come from customer or it can be some intermediate data.

SYSTEMS ALSO EXHIBIT CERTAIN FEATURES AND CHARACTERISTICS:

5. Boundaries

- Every system has defined boundaries within which it operates.
- Personnel system in an organization has its work domain with defined procedures.

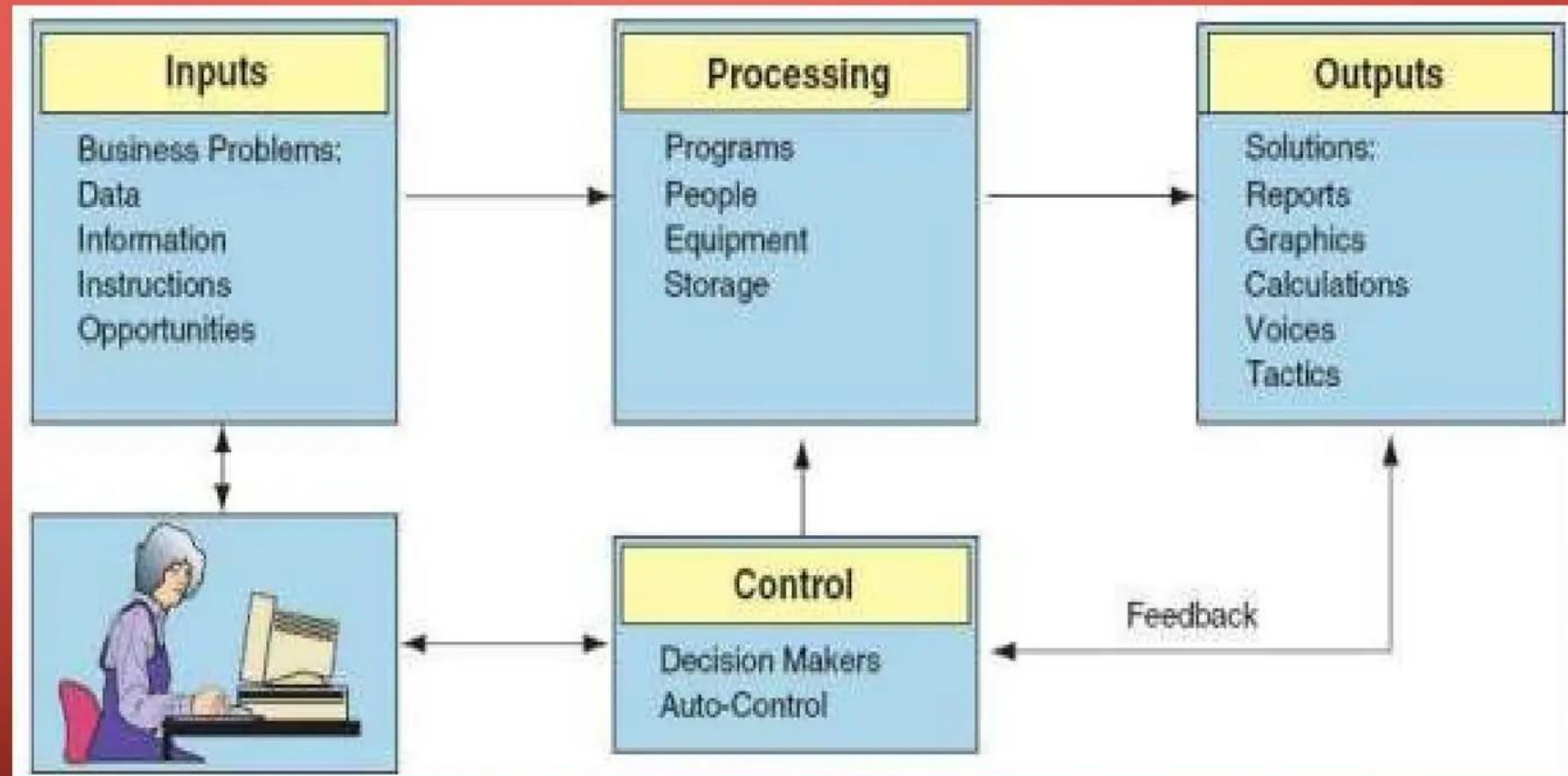
SYSTEMS ALSO EXHIBIT CERTAIN FEATURES AND CHARACTERISTICS:

6. *Interfaces*

- Through interfaces, the system interacts with the outside world.
- These UIs should be as user friendly as possible.

PURPOSE/FUNCTIONS OF AN INFORMATION SYSTEMS

- Inputs
- Processing
- Outputs
- Control



TYPES OF INFORMATION SYSTEMS

1. Office Information Systems
2. Transaction Processing Systems
3. Management Information Systems
4. Decision Support Systems
5. Expert Systems

1. OFFICE INFORMATION SYSTEMS

- OIS refers to the computer machinery and software used to digitally create, collect, store, manipulate, and relay office information.

Main types of tools include:

- MS Word, MS excel etc.
- Text & image processing systems (DTP)
- Personal database systems and note-taking systems

1. OFFICE INFORMATION SYSTEMS

- OIS includes the following roles/activities:
 - Exchange of information
 - Management of administrative documents
 - Handling of numerical data
 - Meeting planning and management of work schedules

1. OFFICE INFORMATION SYSTEMS

Function of OAS

- Create data entry and procedure
 - Generate Documents/Presentations
 - Sends Email
 - Maintain data
 - Create stand-alone apps
- Need/Advantages for office Automation
 - Labor and time saving
 - Accuracy
 - Less chances of committing fraud
 - Uniformity
 - Control of Information

2. TRANSACTION PROCESSING SYSTEMS

- A computerized system that performs and records daily routine transactions necessary to the conduct of the business.
- Usually operated directly by shop floor workers or front line staff, which provide the key data required to support the management of operations.
- This data is usually obtained through the automated or semi automated tracking of low-level activities and basic transactions.

- FIVE STAGES OF TRANSACTION PROCESSING SYSTEM

- Data Entry
 - Processing
 - Database Maintenance
 - Document and Report Generation
 - Inquiry Processing
 - Functions of a TPS
- 

Inputs	Processing	Output
Transactions Events	Validation Sorting Listing Merging Updating Calculation	Lists Detail reports Action reports Summary reports?

Functions of a TPS in terms of data processing requirements

Examples:

- Payroll system
- Reservations systems
- Order processing systems
- Systems for payments and funds transfers
- Billing systems to send invoices to customers
- Systems to calculate the weekly and monthly tax payments

TWO TYPES OF TRANSACTION PROCESSING SYSTEMS

I) BATCH PROCESSING

- Batch processing requires separate programs for input, process and output.
- Batch processing is a resource-saving transaction type that stores data for processing at pre-defined times.
- It is useful for enterprises that need to process large amounts of data using limited resources.

Example

- Credit card's monthly statement for the customer.
- Library books overdue reminders, Journals missing issues reminders.
- NEFT transfer (National Electronic Funds Transfer).

II) REAL TIME PROCESSING

- Involves a continual input, process and output of data.
- Data must be processed in a small time period (or near real time).

Example

- A bank customer withdraws a sum of money from his or her account. The account balance updated as soon as possible.
- Airline ticket booking.
- IMPS (Immediate Payment Service)

FEATURES/CHARACTERISTICS OF TRANSACTION PROCESSING SYSTEMS

1. Rapid Processing

- The rapid processing of transactions is vital to the success of any enterprise.
- TPS systems are designed to process transactions instantly.

2. Reliability

- TPS systems are designed to incorporate comprehensive measures to safeguards and disaster recovery.
- These measures keep the failure rate well within tolerance levels.

3. Standardisation

- Transactions must be processed in the same way each time to maximise efficiency.
- TPS interfaces are designed to acquire identical data for each transaction, regardless of the customer.

4. Controlled Access

- Access must be restricted to authorized employees to use.
- Employees who lack the skills and ability to control, cannot do the transaction process.

3. MANAGEMENT INFORMATION SYSTEMS

- Management Information Systems are management-level systems that are used by middle managers to help ensure the smooth running of the organization in the short to medium term.
- MIS are built on the data provided by the TPS

3. MANAGEMENT INFORMATION SYSTEMS

- The highly structured information provided by these systems allows managers to evaluate an organization's performance by comparing current with previous outputs.
- MIS primarily serve the functions of planning, controlling, and decision making at the management level.

3. MANAGEMENT INFORMATION SYSTEMS

Inputs	Processing	Output
Internal Transactions Internal Files Structured data	Sorting Merging Summarizing	Summary reports Action reports Detailed reports

Functions of a MIS in terms of data processing requirements

Examples of MIS

- Sales management systems
- Inventory control systems
- Budgeting systems
- Management Reporting Systems (MRS)

MIS REPORTS

Several types of reports can be produced

- Summary reports present all activity over a given time period, geographic region, or other categorization in aggregate form.
- Exception reports only present information that is out of normal ranges.
- On-demand reports present anticipated summaries only when a manager wants or needs to check the status of activities.

APPLICATIONS OF MIS

STRATEGY SUPPORT

- Computers cannot create business strategies by themselves but assist management and help in taking effective decision making.
- MIS systems can be used to transform data into information that is useful for decision making.

DATA PROCESSING

- MIS systems provide a valuable time saving benefit to the workforce.
- They allow for faster decision making and quicker reflexes for the enterprise as a whole.

BENEFITS OF MIS

- Core Competency
- Supply Chain Management
- Quick Reflexes

4. DECISION SUPPORT SYSTEMS (DSS)

- Computer-based information system that supports business or organizational decision making activities, typically resulting in ranking, sorting or choosing from among alternatives.
- Knowledge based system, used by senior managers, which facilitates the creation of knowledge and allow its integration into the organization.
- DSS are interactive and are used to solve ill structured problems.
- They offer access to databases, analytical tools, allow "what if" simulations, and support the exchange of information within the organization.

4. DECISION SUPPORT SYSTEMS (DSS)

- DSS usually have three major components:
 - Database
 - Stores large amounts of data relevant to problems the DSS has been designed to tackle.
 - Model base
 - Contains one or more models that can be used to analyze the decision situation
 - Dialogue module
 - Provides a way for the decision maker, usually a non-technical manager, to communicate with the DSS.

DSS MANIPULATE AND BUILD UPON THE INFORMATION FROM A MIS AND/OR TPS TO GENERATE INSIGHTS AND NEW INFORMATION.

Functions of a DSS in terms of data processing requirements

Inputs	Processing	Output
Internal Transactions Internal Files External Information?	Modeling Simulation Analysis Summarizing	Summary reports Forecasts Graphs/Plots

Examples of DSS

- Group Decision Support Systems (GDSS)
- Computer Supported Co-operative work (CSCW)
- Logistics systems
- Financial Planning systems
- Improve in overall efficiency
- Competitive advantage
- Better Satisfaction

5. EXPERT SYSTEMS

- Support professionals faced with complex situations requiring expert knowledge in a well-defined area.
- Expert systems are based on principles of artificial intelligence research.
- These systems use human knowledge captured in a computer to solve problems that ordinarily need human expertise. Mimicking human expertise and intelligence requires that the computer
 - recognize, formulate and solve a problem;
 - explain solutions and
 - learn from experience.

These systems explain the logic of their advice to the user; hence, in addition to solving problems they can also serve as a teacher.

They use flexible thinking processes and can accommodate new knowledge.

5. EXPERT SYSTEMS

- Expert systems are based on probabilities computers can be programmed with “subject knowledge” to mimic the role of experts.
- There are two primary parts to expert systems
 - Knowledge base
 - The knowledge base contains both factual and judgmental knowledge.
 - Reasoning engine.
 - The reasoning engine uses inference to solve problems, often using "if-then" decision chains.

5. EXPERT SYSTEMS

Example:

- Artificial intelligence system that emulates an auto mechanic's knowledge in diagnosing automobile problems. This hypothetical expert system would likely be the result of engineering using an actual mechanic's knowledge base.
- One of the most common uses of expert systems is in medicine
 - The ONCOLOG system shown here analyses patient data to provide a reference for doctors, and help for the choice, prescription and follow-up of chemotherapy